

Chapter 3.

3.1. (a) $x_1 = 1, x_2 = 2, x_3 = 1$

(b) $x_1 = 2, x_2 = -\frac{5}{3}, x_3 = -\frac{4}{3}$

3.2. (a) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 4 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 6 \end{bmatrix}$

(b) $L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -20 \end{bmatrix}$

(c) $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ -3 & -5 & -\frac{22}{9} & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & 2 & -4 \\ 0 & 0 & -9 & 1 \\ 0 & 0 & 0 & -\frac{68}{9} \end{bmatrix}$

3.3. (a) $x = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

(b) $x = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$

3.4. Let $A = (a_{ij})_{n \times n}$, $B = (b_{ij})_{n \times n}$ be two $n \times n$ matrices and $C = AB = (c_{ij})_{n \times n}$

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = \underbrace{a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}}_{n \text{ products} + (n-1) \text{ sums.}}$$

requires $2n-1$ flops.

Each component of C (totally n^2) has the same complexity, so the total cost is $n^2(2n-1) = 2n^3 - n^2$ flops.

3.5. $\frac{n(n+1)}{2} + \frac{(n-1)n}{2} = \frac{(n+1+n-1) \cdot n}{2} = n^2$
 $\downarrow$ for products $\downarrow$ for sums.

3.6. $(uv^t)w$: cost $3n^2 - n$ flops
 $u(v^tw)$: cost $2n-1$ flops $\left. \vphantom{\begin{matrix} (uv^t)w \\ u(v^tw) \end{matrix}} \right\} u(v^tw)$ is more efficient

3.7. Do it yourself.

3.8. Proof: We need to show.

$$\sum_{k=1}^n k^m \leq C n^{m+1} \text{ for some constant } C \text{ that does not depend on } n.$$

Since $k \leq n$, it is true $k^m \leq n^m$ for $k=1, 2, \dots, n$.

Thus

$$\sum_{k=1}^n k^m = 1^m + 2^m + 3^m + \dots + n^m \leq \underbrace{n^m + n^m + n^m + \dots + n^m}_{n \text{ terms}} \\ = n \cdot n^m = n^{m+1}$$

(i.e., we can choose $C=1$ if desired).